03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 19

(JCD 13)

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Previous lecture

- Low-mass stars have no convective cores, so H exhausts gradually
- H shell first thick, then thin
- Core becomes degenerate prior to He fusion
- This leads to the He flash after which the star settles back down to equilibrium
- Subsequent evolution is the same as for moderate-mass stars

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Outline

Isochrones versus evolutionary tracks

Stellar clusters

Observing stellar clusters

Inferred parameters from observing stellar clusters

Summary

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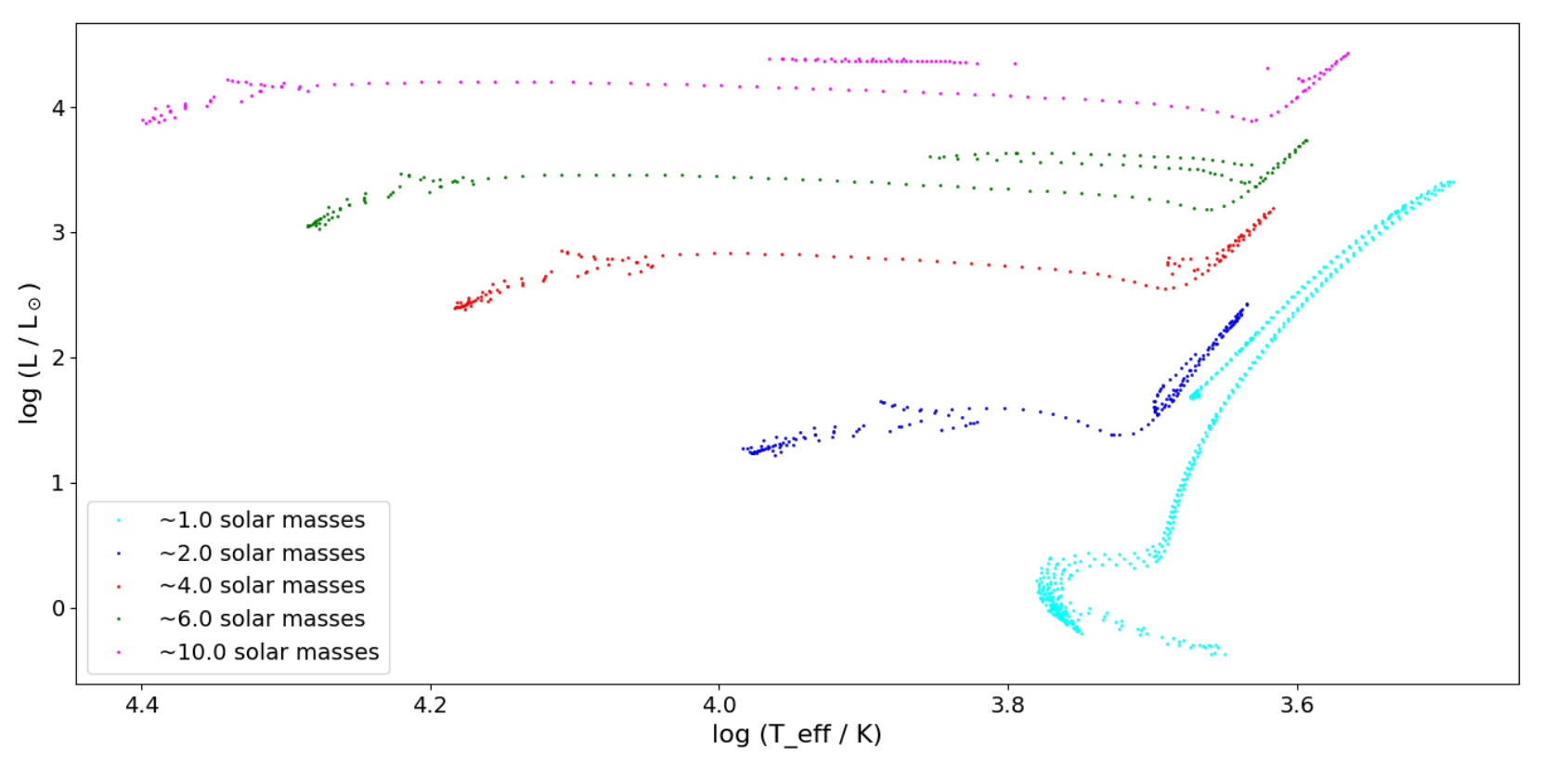
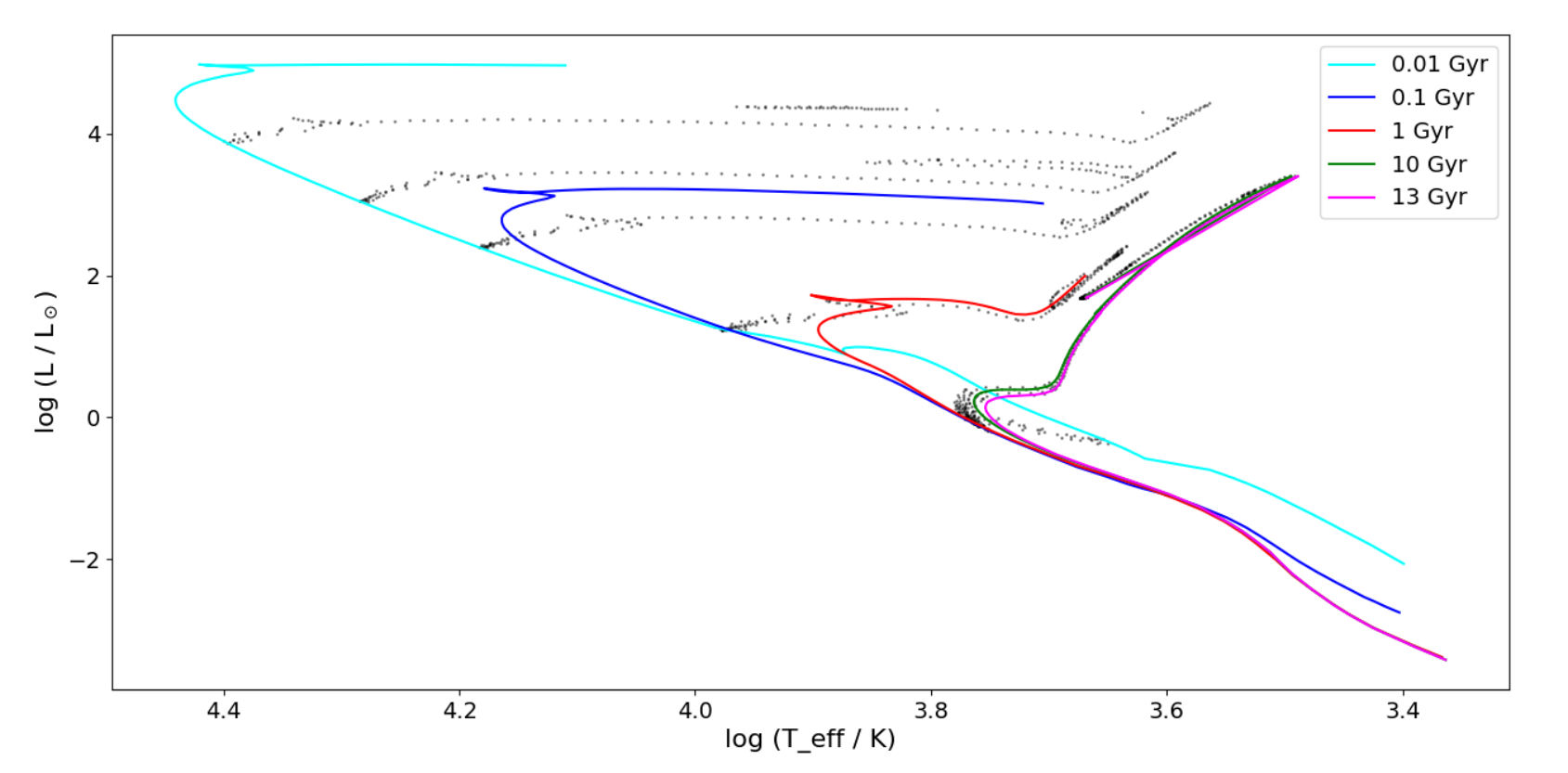
Evolutionary tracks versus isochrones

- A star's position in the HR diagram can be fully determined by its mass, age and chemical composition.
  - Note that observationally, you would additionally need the stellar distance and the interstellar extinction
- Evolutionary tracks: curve in HR diagram for a fixed mass and chemical composition and with varying age
  - Note that these are the ones we have been looking at in lectures till now
- Isochrones: curve in HR diagram for a fixed age and chemical composition but for varying mass
- As a result of faster evolution after the MS, evolutionary tracks and isochrones are very similar.

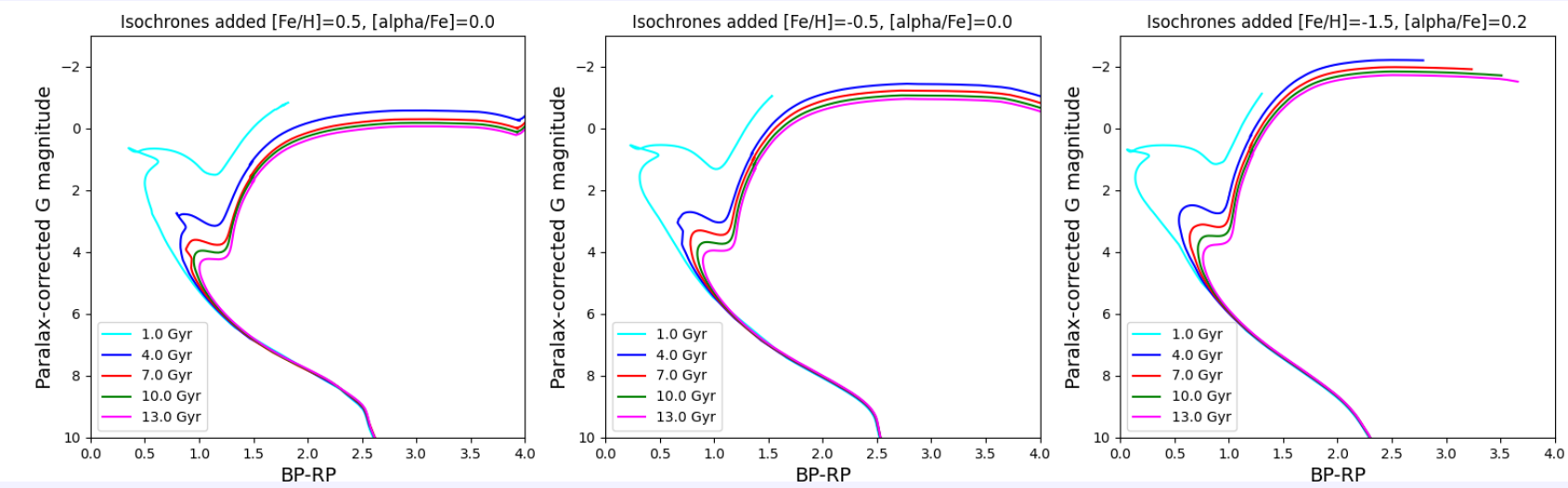
PARSEC Evolutionary tracks



PARSEC Evolutionary tracks + isochrones



Isochrone dependence on chemical composition



- Differences mainly come from X (hydrogen mass fraction) being larger for low-metallicity stars

Turn-off mass

- For a single isochrone, the lowest-mass stars are still on the main sequence while the more massive stars can/will be further in their evolution
- This is a result of evolution timescales being dependent on stellar mass
- The mass of the star where the curve transitions away from the main sequence is called the "turn-off mass", M_t

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Stellar clusters

- Stars that are gravitationally bound together
- Two types: globular and open
- Assumed to have formed from the same interstellar gas
- Thus assumed to have approximately the same age and chemical composition
- Hence stars roughly follow a single isochrone in the HR diagram

Globular and open clusters

Left: M2 cluster in Aquarius. Right: NGC 457 (ET cluster) in Cassiopeia.

(Credit: Sky & telescope, Hunter Wilson Astrophotography)

Globular clusters

- Distribution: galactic halo + nucleus
- Chemical composition: metal-poor $Z \sim 0.0001 - 0.001$
- Ages: Very old - several 10^{10} years
- Density: High - $10^4 - 10^6$ stars in $20 - 100\,\mathrm{pc}$

Open clusters

- Distribution: Milky Way plane / Galactic disc
- Chemical composition: metal-rich $Z \sim 0.01 - 0.04$
- Ages: young to old - $10^6 - 10^9$ years
- Density: Low - $< 10^3$ stars in 10 pc

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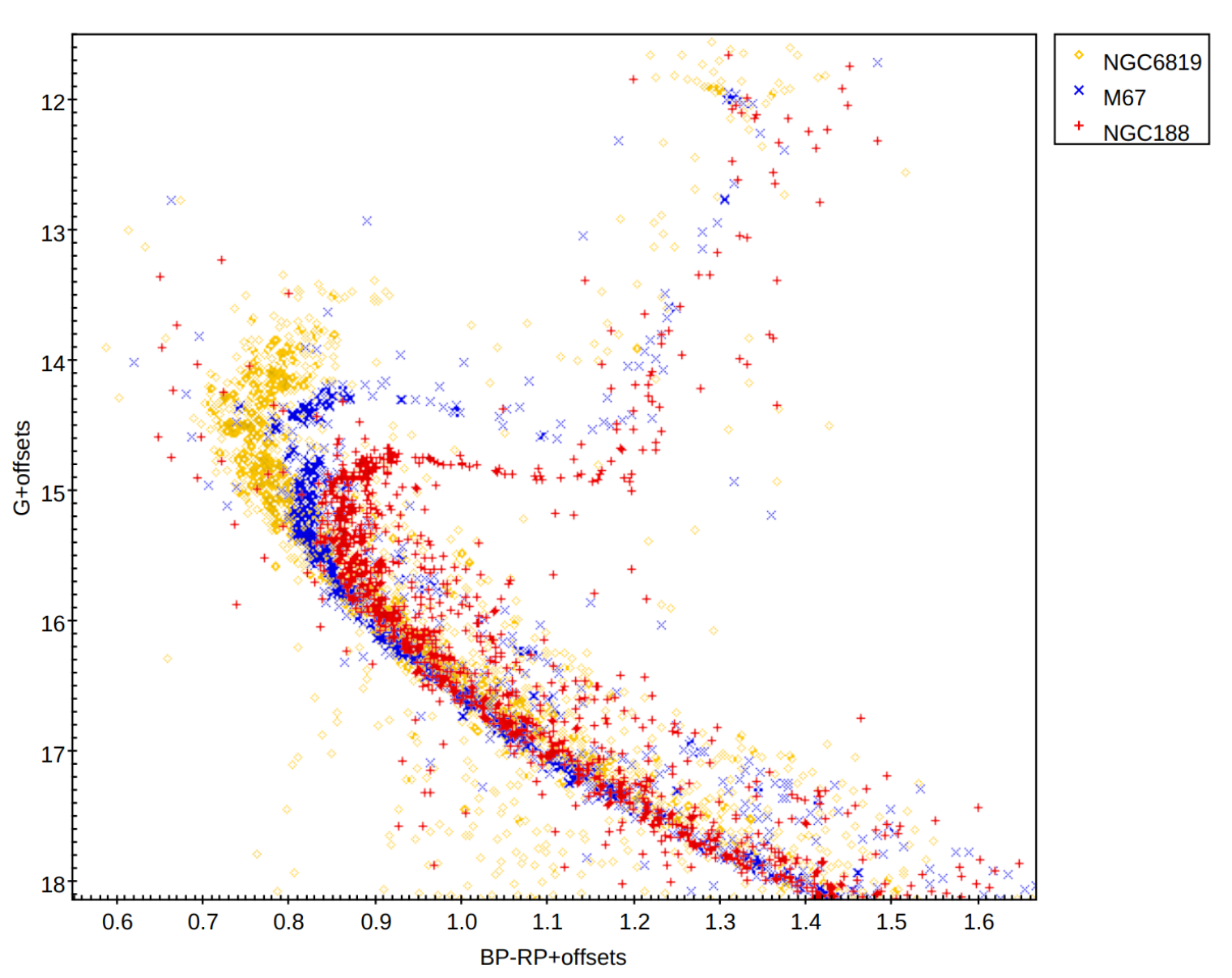
Observed HR diagram for open clusters



(Credit: Carrasco et al. 2021 / JCD - Fig 13.6)

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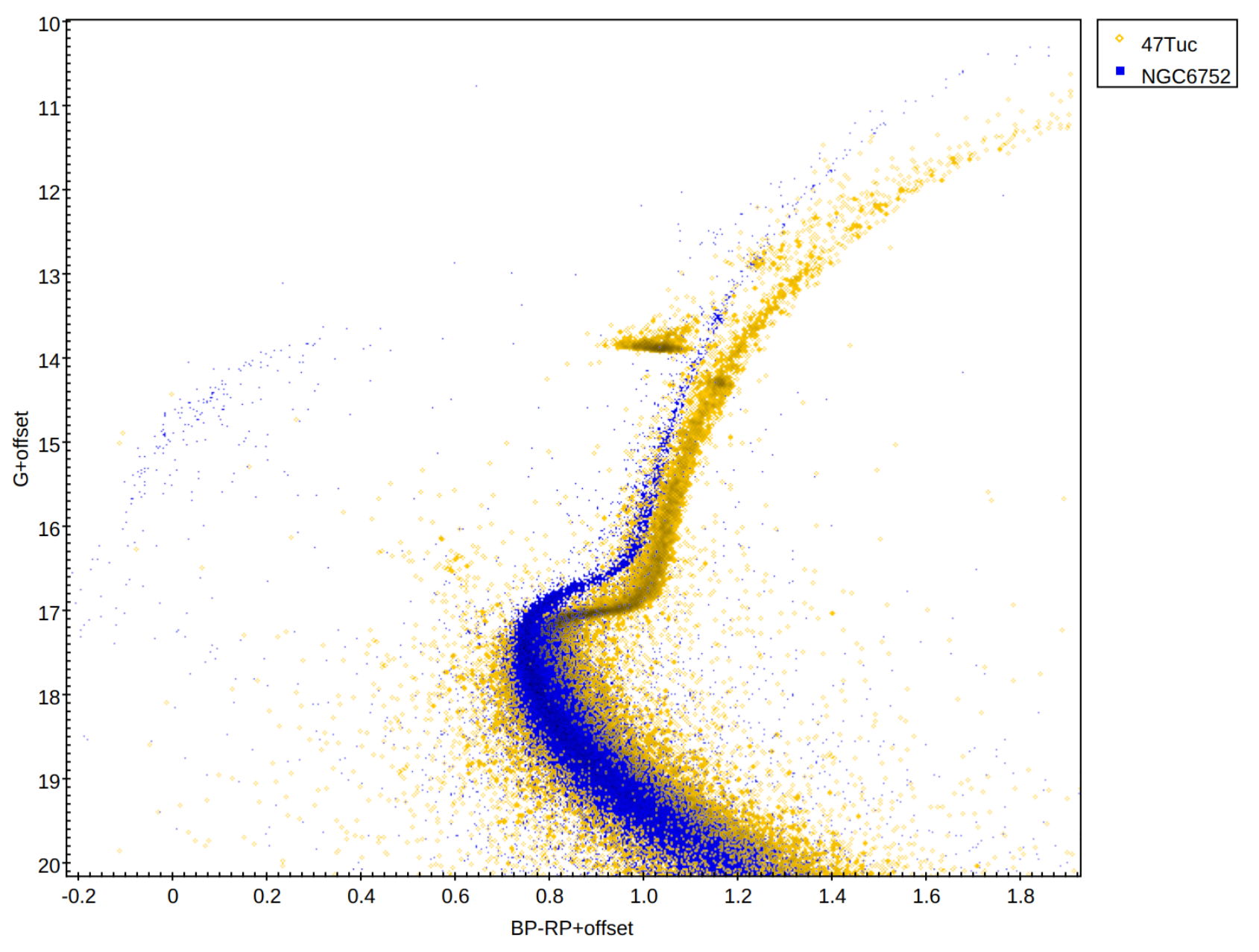
Observed HR diagram for globular clusters



(Credit: Carrasco et al. 2021 / JCD - Fig 13.7)

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Distances

- We can observe a star's effective temperature (or colour) and apparent luminosity (or apparent magnitude)
- In the HR diagram, we need absolute luminosity (or absolute magnitude)
- Remember from lecture 3: these are related by the distance to the star
- So, from "sliding" the observed isochrone of a cluster in the HR diagram to match the location of the main sequence, we can measure the distance to the cluster since the "displacement" is directly related to distance
  - Note that astrometry and in particular the Gaia mission can measure distances independently for all stars

Ages

- Once the cluster distance is known and the cluster is in the correct place in the HR diagram, one can use the turn-off point to measure the age of the cluster
- We don't know the actual turn-off mass, but we know the effective temperature and luminosity at the turn-off point
- Via either MS relations or better, by fitting theoretical isochrones, we can now determine the age of the cluster
- Note that this still relies on chemical composition, but that can be determined via spectroscopic measurements

Identifying unresolved binaries

- If a cluster has unresolved binaries, they are displaced upwards in the HR diagram
- This is due to the colour being an average of the two components' colours and thus close to the value of a single star, but the luminosity is the sum of both components and thus the luminosity / absolute magnitude will be higher

Summary

- Evolutionary tracks: curve in HR diagram for a fixed mass and chemical composition and with varying age
- Isochrones: curve in HR diagram for a fixed age and chemical composition but for varying mass
- Stellar clusters lie along an isochrone and observations of clusters can thus measure precise ages

NEXT LECTURE: Supermassive star evolution